Are the two $(1+t)$'s the same $(1+t)$?

Q105: a separator for the ribbon-side and vertex-side deformations

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Title: Are the two $(1+t)$'s the same $(1+t)$? Q105: a separator for the ribbon-side and vertex-side deformations
Corresponds to: `clio-vega/proofs@69f93c6` (paper, code, and the seven `Q105-*` nodes of `registry/fock-ribbon-sign-operator.json`)
Builds on: `clio-vega/proofs@0064531` (Q99) and the Q92/Q96 papers in the same repository.
Not built on: `grandpa-rick/rick-research@616ea6e` §4, which §5.5 below refutes as stated.

Verdict: DISTINCT.
The order of vanishing at $t=-1$ separates them, and it is not a fitted quantity on either side. The ribbon-side order function is the *constant* 1; the vertex-side order function is $(m+n)$ mod 2 and attains 0. A constant cannot be identified with a free coordinate. The alphabet question is answered **(ii)**: the published twist alphabet $(1-t)$ and my defect alphabet $(1+t)$ are functions of *different data* — the twist is (zero − pole) of the contraction, the defect is $1+$ (pole) — and unpinning the zero moves one while fixing the other.

Abstract

Two objects produced on 8 September 2026 each carry a factor $(1+t)$, and both vanish at $t=-1$ in the crude sense that made the coincidence tempting. **(A)** is the one-bead sector of the cross-rank ribbon commutator $[R_e(t), R_f(s)]$ restricted to the hyperbola $ts=1$; **(B)** is the multiplication operator $h_{m+n}[(1+t)X]$ that measures the failure of the Hall–Littlewood vertex-operator modes to generate a quantum torus on the same hyperbola. We settle whether they are one object. On the ribbon side we obtain the closed form $\langle M'|[R_e(t), R_f(t^{-1})]|M\rangle = (-1)^N(-t)^P\big(1-(-t)^\kappa\big)$ for a single integer κ read off the Maya diagram, from which the matrix element is zero iff $\kappa=0$ and otherwise has order of vanishing *exactly* 1 at $t=-1$, uniformly in e, f and in the bead configuration. On the vertex side the matrix elements of $h_N[(1+t)X]$ are palindromic, hence their order of vanishing at $t=-1$ is congruent to N mod 2; for N even it is never 1 and is 0 on an explicit family. The two order functions therefore disagree, and no dictionary between the parameter sets repairs them. Separately we locate the $(1+t)$ of (B): it is not a Wick constant. The contraction constant of the vertex operators is $(z_1-z_2)/(z_1-tz_2)$, with twist alphabet $(1-t)$ exactly as published by Necoechea–Rozhkovskaya; its *pole* sits at $z_1 = tz_2$, and $(1+t)X$ is the doubled *creation* alphabet $\sigma[Xz_1]\sigma[Xz_2]$ evaluated at that pole. Unpinning the zero of the contraction — a parameter forced to 1 by the Hall–Littlewood normalisation — moves the twist alphabet while leaving $(1+t)$ fixed, which is a varying test that no shape match survives.

1 The two objects, and what would count as an answer

Conventions. Λ is the ring of symmetric functions over $\mathbb{Q}(t,s)$. For a partition λ the Maya set is $M(\lambda) = \{\lambda_j - j : j \geq 1\} \subset \mathbb{Z}$, its elements *beads*; for $b \in M$ and $g \geq 1$ with $b + g \notin M$ we call (b,g) a legal g-move and write $M \cdot (b,g) = (M \setminus \{b\}) \cup \{b+g\}$. The operator $R_g(u)$ adds a connected g-ribbon with weight u^{ht}; in Maya form $R_g(u)|M\rangle = \sum_{(b,g)\text{ legal}} u^{\#(M\cap(b,b+g))}|M\cdot(b,g)\rangle$ [1, Prop. 2.2]. The Hall–Littlewood vertex operator is $H_u(z)f[X] = \sigma[Xz]\, f[X-(1-u)/z]$ with modes H^u_m [3, Def. 1.1]. For a nonzero $F \in \mathbb{Q}[t^{\pm}]$ (or $F \in \Lambda \otimes \mathbb{Q}[t^{\pm}]$) we write $\operatorname{ord} F$ for the order of vanishing at $t=-1$; for a symmetric function we take the minimum over the coefficients in a fixed basis, which is basis-independent.

(A), the ribbon-side $(1+t)$. On $ts = 1$ the two-bead sector of $[R_e(t), R_f(s)]$ vanishes identically [2, Cor. 4.2(iii)], so the commutator is carried entirely by its one-bead sector, whose matrix elements are divisible by $(1+t)$.

(B), the vertex-side $(1+t)$. For $s = t^{-1}$,

$$H^t_m H^{1/t}_n - t^{-1} H^{1/t}_n H^t_m = \big(1 - t^{-1}\big) t^{-m}\, h_{m+n}\big[(1+t)X\big]\cdot, \tag{1}$$

a multiplication operator, the plethysm being $p_k \mapsto (1+t^k)p_k$ [3, Thm. 3.4].

The trap. Both objects vanish at $t=-1$ in *some* instances, and $(1+t^k)$ is the Hall–Littlewood/Schur-Q doubling deformation, so a common origin is plausible. That is a shape match. Agreement of values at $t=-1$ is worthless here: it is a check with a kernel in exactly the direction it is meant to test. We therefore fix in advance the two coordinates that decide the question, neither of which can be tuned:

(S1) the *order* of vanishing at $t=-1$, as a function on each side's own parameter space (not the value, and not the fact of vanishing);

(S2) the origin of the alphabet: is $(1+t)$ a contraction (Wick) constant of a t-twisted fermion, or something else?

A merge and a split are equally good outcomes. "Both have a $(1+t)$" is not an outcome.

2 The ribbon side: one integer controls everything

We start from the two-parameter matrix elements, which are already proved:

Proposition 2.1 ([2, Thm. 4.1(1)])**.** *Let $e, f \geq 1$, let M be a Maya set and let $M' = M\cdot(a, e+f)$ be a legal $(e+f)$-move. Put*

$$N = \#\big(M\cap(a,a+e+f)\big), \quad A = \#\big(M\cap(a,a+f)\big), \quad B = \#\big(M\cap(a,a+e)\big),$$

$m_e = \mathbf{1}[a+e\in M]$ *and* $m_f = \mathbf{1}[a+f\in M]$*. Then*

$$\langle M'|[R_e(t),R_f(s)]|M\rangle = s^A t^{N-1-A}\big(t - m_f(1+t)\big) - t^B s^{N-1-B}\big(s - m_e(1+s)\big).$$

Definition 2.2. With the notation of Proposition 2.1 set

$$\boxed{\kappa = \kappa^{e,f}_a(M) := 2\big(A+B-N\big) + m_e + m_f}, \qquad P := N - 2A - m_f.$$

κ is an integer attached to the bead configuration; it is not a fitted parameter and it is not constant — on the sweep of §6 it takes every value in $\{-4, \dots, 6\}$.

Theorem 2.3 (the hyperbola closed form)**.** *Let $e, f \geq 1$ and $s = t^{-1}$. Then for every Maya set M and every legal $(e{+}f)$-move $(a, e+f)$, with $M' = M \cdot (a, e+f)$,*

$$\boxed{\langle M' | [R_e(t), R_f(t^{-1})] | M \rangle \;=\; (-1)^N \, (-t)^P \, \bigl(1 - (-t)^\kappa\bigr).}$$

Consequently:

(i) the matrix element is zero if and only if $\kappa = 0$;

*(ii) if $\kappa \neq 0$ its order of vanishing at $t = -1$ is **exactly** 1, with*

$$\langle M' | [R_e(t), R_f(t^{-1})] | M \rangle \;\equiv\; (-1)^N \kappa \, (t+1) \quad (\mathrm{mod}\ (t+1)^2);$$

(iii) in particular $(1+t)$ divides every one-bead matrix element, uniformly in e, f and in M, and divides no such element twice.

Proof. Since $m_f \in \{0, 1\}$ we have $t - m_f(1+t) = t$ when $m_f = 0$ and $= -1$ when $m_f = 1$, i.e.

$$t - m_f(1+t) = (-1)^{m_f} \, t^{1-m_f}.$$

Likewise $s - m_e(1+s) = (-1)^{m_e} s^{1-m_e}$, which at $s = t^{-1}$ is $(-1)^{m_e} t^{m_e - 1}$. Substituting $s = t^{-1}$ into Proposition 2.1,

$$s^A t^{N-1-A} = t^{N-1-2A}, \qquad t^B s^{N-1-B} = t^{2B-N+1},$$

so the matrix element equals

$$\Xi := (-1)^{m_f} \, t^{N-2A-m_f} \;-\; (-1)^{m_e} \, t^{2B-N+m_e} \;=\; (-1)^{m_f} t^P - (-1)^{m_e} t^Q, \tag{2}$$

where $P = N - 2A - m_f$ and $Q := 2B - N + m_e$. Note

$$Q - P = 2B - N + m_e - N + 2A + m_f = 2(A + B - N) + m_e + m_f = \kappa.$$

Now the two signs collapse. Since $P \equiv N + m_f$ and $Q \equiv N + m_e \pmod 2$,

$$(-1)^{m_f} = (-1)^{N+P}, \qquad (-1)^{m_e} = (-1)^{N+Q},$$

so (2) becomes

$$\Xi = (-1)^N \bigl[(-1)^P t^P - (-1)^Q t^Q\bigr] = (-1)^N \bigl[(-t)^P - (-t)^Q\bigr] = (-1)^N (-t)^P \bigl(1 - (-t)^\kappa\bigr),$$

using $Q = P + \kappa$. This is the closed form.

For (i): $(-t)^P$ is a unit in $\mathbb{Q}[t^{\pm}]$, so $\Xi = 0$ iff $(-t)^\kappa = 1$ in $\mathbb{Q}[t^{\pm}]$, iff $\kappa = 0$. For (ii): substituting $t = -1$ gives $(-t) = 1$ and $1 - 1^\kappa = 0$, so $(t{+}1) \mid \Xi$ always. Differentiating, $\frac{d}{dt}\bigl(1 - (-t)^\kappa\bigr) = \kappa (-t)^{\kappa - 1}$, which at $t = -1$ equals κ; and $(-1)^N (-t)^P$ equals $(-1)^N$ at $t = -1$. Hence $\Xi'(-1) = (-1)^N \kappa$, nonzero precisely when $\kappa \neq 0$, which is (ii); and (iii) follows. $\square$

Remark 2.4 (what the $(1+t)$ *is*, on this side)**.** Theorem 2.3 exhibits the ribbon-side $(1+t)$ as the first cyclotomic factor of a quantum integer at $q=-t$: for $\kappa>0$,

$$1-(-t)^{\kappa}=\bigl(1-(-t)\bigr)\sum_{j=0}^{\kappa-1}(-t)^{j}=(1+t)\,[\kappa]_{-t}.$$

It is a *divisor*, and the exponent κ is a bead statistic. This is the same shape as the two-bead sector, whose matrix elements carry the factor $1-(ts)^{k}$ [2, Thm. 4.1(2)] and which is what makes that sector vanish on $ts=1$; the one-bead sector is the same mechanism with ts replaced by $-t$ and the interference index k replaced by κ. Nothing here is an alphabet, and no companion factor $(1+t^{j})$ with $j\geq 2$ ever appears.

Corollary 2.5. *On the hyperbola $ts=1$ the operator $[R_e(t),R_f(t^{-1})]$ has bead-number exactly* 1*, and the order function*

$$\omega_A:\ (e,f,M,M')\ \longmapsto\ \operatorname{ord}\,\langle M'|[R_e(t),R_f(t^{-1})]|M\rangle\ \in\ \{1\}$$

is ***constant*** *on the whole of its (nonvanishing) domain.*

Proof. The two-bead sector vanishes identically on $ts=1$ because each of its summands carries the factor $1-(ts)^{k}$ [2, Thm. 4.1(2), Cor. 4.2(iii)], and sectors of bead-number ≥ 3 do not occur since each R_g has bead-number 1 [1, Conv. 2.1]. The constancy is Theorem 2.3(ii). $\square$

3 The vertex side: palindromy fixes the parity of the order

Lemma 3.1 (the h-expansion)**.** *For $N\geq 0$,* $\quad h_N\bigl[(1+t)X\bigr]=\sum_{k=0}^{N}t^{N-k}\,h_k\,h_{N-k}$*.*

Proof. $\sigma[(1+t)Xz]=\sigma[Xz]\,\sigma[tXz]$ because $p_n[(1+t)X]=p_n[X]+p_n[tX]$ and $\sigma[A+B]=\sigma[A]\sigma[B]$. Comparing coefficients of z^N and using $h_j[tX]=t^j h_j$ gives the claim. $\square$

Lemma 3.2 (palindromy)**.** *Fix partitions λ,ν with $|\nu|=|\lambda|+N$, and set $c_k:=\bigl\langle s_\nu,\ h_kh_{N-k}\,s_\lambda\bigr\rangle\in\mathbb{Z}_{\geq 0}$ and $\Pi_{\nu\lambda}(t):=\bigl\langle s_\nu,\ h_N[(1+t)X]\,s_\lambda\bigr\rangle$. Then $\Pi_{\nu\lambda}(t)=\sum_{k=0}^{N}c_kt^{N-k}$ and $c_k=c_{N-k}$, i.e. $\Pi_{\nu\lambda}(t)=t^N\Pi_{\nu\lambda}(1/t)$.*

Proof. The displayed expansion is Lemma 3.1 paired against s_ν. The symmetry is $h_kh_{N-k}=h_{N-k}h_k$, which gives $c_k=c_{N-k}$ on the nose. $\square$

Lemma 3.3 (order parity of a self-reciprocal polynomial)**.** *Let $0\neq\Pi\in\mathbb{Q}[t]$ satisfy $\Pi(t)=t^N\Pi(1/t)$. Then the multiplicity of $t=-1$ as a root of Π is congruent to N modulo* 2*.*

Proof. Write $\Pi=t^{\alpha}Q$ with $Q(0)\neq 0$. From $\Pi(t)=t^N\Pi(1/t)$ we get $Q(t)=t^{N-2\alpha}Q(1/t)$, and since $Q(0)\neq 0$ this forces $\deg Q=N-2\alpha$. Factor $Q=(t-1)^{m_+}(t+1)^{m_-}R$ with $R(\pm 1)\neq 0$. The nonreal-or-non-unit roots of Q occur in pairs $\{\rho,1/\rho\}$ with equal multiplicities and $\rho\neq\pm 1$, so $\deg R$ is even and $\deg Q\equiv m_++m_-\pmod 2$. Finally m_+ is even: writing $Q=(t-1)^{m_+}S$ with $S(1)\neq 0$, self-reciprocity of Q forces $S=(-1)^{m_+}S^*$ where $S^*(t)=t^{\deg S}S(1/t)$; if m_+ were odd this would give $S(1)=-S^*(1)=-S(1)$, hence $S(1)=0$, a contradiction. So $m_-\equiv\deg Q=N-2\alpha\equiv N\pmod 2$. $\square$

Theorem 3.4 (the vertex-side order function)**.** *Let $N=m+n\geq 0$ and let $D_{m,n}(t)=(1-t^{-1})t^{-m}h_N[(1+t)X]$ be the defect operator of* (1)*. Then:*

(i) ord $D_{m,n} = $ ord $h_N[(1+t)X] = N \bmod 2$*, as an element of* $\Lambda \otimes \mathbb{Q}[t^{\pm}]$*;*

(ii) every nonzero Schur matrix element of $D_{m,n}$ *has order of vanishing* $\equiv N \pmod 2$*; in particular for* N even *no matrix element has order* 1*;*

(iii) for N *even the order* 0 *is attained: with* $\lambda = \varnothing$ *and* $\nu = (N)$ *one has* $\Pi_{\nu\lambda}(t) = 1+t+\cdots+t^N$*, whose value at* $t=-1$ *is* 1*.*

Hence the order function $\omega_B : (\lambda, \nu, m, n) \mapsto \operatorname{ord}\langle s_\nu | D_{m,n} | s_\lambda\rangle$ *takes both parities and attains the value* 0*.*

Proof. (i) The prefactor $(1-t^{-1})$ takes the value 2 at $t=-1$ and t^{-m} is a unit, so neither changes the order. In the power-sum basis $h_N[(1+t)X] = \sum_{\mu \vdash N} z_\mu^{-1} \prod_i (1+t^{\mu_i}) p_\mu$, and $(1+t^j)$ vanishes at $t=-1$ exactly when j is odd, to order 1. So the coefficient of p_μ has order equal to the number $o(\mu)$ of odd parts of μ, and the order of the whole is $\min_{\mu \vdash N} o(\mu)$. Since $o(\mu) \equiv |\mu| = N \pmod 2$, this minimum is 0 for N even (take $\mu = (2^{N/2})$) and 1 for N odd (take $\mu = (2^{(N-1)/2}, 1)$).

(ii) Combine Lemmas 3.2 and 3.3.

(iii) $\langle s_{(N)}, h_k h_{N-k}\rangle = K_{(N),(k,N-k)} = 1$ for every k, so $c_k = 1$ for all k and $\Pi = \sum_{k=0}^N t^k$; at $t=-1$ this alternating sum is 1 when N is even. ☐

Remark 3.5 (a ladder, not a single slice)**.** The proof of (i) shows more: the per-partition orders form the full ladder $\{N \bmod 2,\ N \bmod 2 + 2,\ \ldots,\ N\}$, and the Schur matrix elements realise the same ladder (computed in §6). So on the vertex side the order at $t=-1$ is a genuine *grading* by the number of odd parts. On the ribbon side there is no ladder: the order is 1 or the element is zero.

4 The decision

Theorem 4.1 (Q105)**.** *The objects (A) and (B) are* ***distinct****. Precisely: there is no map* ϕ *from the parameter set of (A) to that of (B) with* $\omega_A = \omega_B \circ \phi$ *and* ϕ *surjective onto the parameter* $m+n$ *of (B).*

Proof. By Corollary 2.5, $\omega_A \equiv 1$. By Theorem 3.4(ii), $\omega_B \equiv m+n \pmod 2$. If $\omega_A = \omega_B \circ \phi$ then $m+n$ is odd at every point of the image of ϕ; but $D_{m,n} \neq 0$ for every $m+n \geq 0$ (its $\lambda = \varnothing$, $\nu = (m+n)$ matrix element is $(1-t^{-1})t^{-m}[m+n+1]_t \neq 0$), so the even-$(m+n)$ half of (B) is a nonempty family of nonzero operators outside the image. Hence ϕ is not surjective. ☐

Remark 4.2 (what the shape match confused)**.** The two $(1+t)$'s play different grammatical roles.

	(A) ribbon	**(B) vertex**
role of $(1+t)$	a *divisor* of the object	a *plethystic alphabet* $X \mapsto (1+t)X$
divides the object?	always	only when $m+n$ is odd
order at $t=-1$	exactly 1, always	$\equiv m+n \pmod 2$, ladder up to $m+n$
higher modes $(1+t^j)$, $j \geq 2$	never occur	all occur
bead-number of the operator	exactly 1	unbounded (multiplication)

"Divisible by $(1+t)$" and "obtained by the plethysm $(1+t)X$" are different predicates that agree only in odd degree. The tempting inference identified a factor with an argument.

Remark 4.3 (the separator would have fired the other way too)**.** Nothing above was tuned to produce a split. Had ω_A come out equal to $e+f \bmod 2$, the dictionary $m+n = e+f$ would have matched the two order functions exactly and the honest report would have been a merge. It came out constant.

5 The alphabet: (1-t) is the twist, (1+t) is the pole

The remaining question is (S2): where does the $(1+t)$ of (1) come from? The published twist alphabet for the Hall–Littlewood boson–fermion correspondence is $(1-t)$, not $(1+t)$: Necoechea–Rozhkovskaya obtain the Hall–Littlewood presentation from the Schur one "by a simple twist of the fields of charged free fermions by multiplication by $E(-u/t)$ or $H(u/t)$", with $\Psi^+(u) = E(-u/t)\Phi^+(u)$ and $\Psi^-(u) = H(u/t)\Phi^-(u)$; the twisted fields satisfy an anticommutation relation whose right-hand side is $\delta(u,v)(1-t)^2$, and $H(u)E(-u/t) = \exp\bigl(\sum_{n\geq 1} \frac{(1-t^n)p_n}{n} u^{-n}\bigr)$ [4, ll. 385, 392–394, 398–405, 735, 764]. Note $t \mapsto -t$ does *not* carry $1-t^n$ to $1+t^n$: it does so only for odd n. So either contracting a $(1-t)$-twisted field against a $(1-t^{-1})$-twisted one produces $(1+t)$ after simplification, or the two alphabets are genuinely different. We show the latter, and identify what $(1+t)$ actually is.

Definition 5.1. For parameters a, b put $\widetilde{H}_{a,b}(z)f[X] = \sigma[Xz]\ f\left[X - \dfrac{a-b}{z}\right]$, the shift being the plethystic substitution $p_n \mapsto p_n - (a^n - b^n)z^{-n}$. The Hall–Littlewood operator is the pin $a = 1$: $H_u = \widetilde{H}_{1,u}$.

Proposition 5.2 (zero and pole)**.** *Let Φ' denote the normal-ordered series $\Phi' = \sigma[Xz_1]\sigma[Xz_2]\ f\bigl[X - \frac{a-b}{z_1} - \frac{c-d}{z_2}\bigr]$. Then*

$$(z_1 - bz_2)\,\widetilde{H}_{a,b}(z_1)\widetilde{H}_{c,d}(z_2) = (z_1 - az_2)\,\Phi', \qquad (z_2 - dz_1)\,\widetilde{H}_{c,d}(z_2)\widetilde{H}_{a,b}(z_1) = (z_2 - cz_1)\,\Phi'.$$

In particular the contraction constant of $\widetilde{H}_{a,b}(z_1)$ against $\widetilde{H}_{c,d}(z_2)$ is $\dfrac{z_1 - az_2}{z_1 - bz_2}$: it has its zero *at $z_1 = az_2$ and its* pole *at $z_1 = bz_2$, and its alphabet is $(a-b)$.*

Proof. The only contraction is between the shift of the left operator and the $\sigma[Xz_2]$ of the right one. Substituting $X \mapsto X - \frac{a-b}{z_1}$ into $\sigma[Xz_2]$ gives $\sigma[Xz_2] \cdot \sigma\bigl[-(a-b)\frac{z_2}{z_1}\bigr]$, and

$$\sigma\Bigl[-(a-b)\tfrac{z_2}{z_1}\Bigr] = \exp\Bigl(-\sum_{n\geq 1} \tfrac{a^n - b^n}{n}\bigl(\tfrac{z_2}{z_1}\bigr)^n\Bigr) = \frac{1 - az_2/z_1}{1 - bz_2/z_1} = \frac{z_1 - az_2}{z_1 - bz_2},$$

the middle step being $-\sum_n \frac{(a^n - b^n)y^n}{n} = \log(1-ay) - \log(1-by)$. Moving the shift of the left operator to the right of $\sigma[Xz_2]$ therefore produces exactly this factor times Φ', which is the first identity; the second is the same computation with the roles of the two operators exchanged. (At $a = c = 1$, $b = t$, $d = s$ the two identities give $(z_1 - tz_2)H_t(z_1)H_s(z_2) = (sz_1 - z_2)H_s(z_2)H_t(z_1)$, which is [3, Thm. 2.4].) $\square$

Theorem 5.3 (the defect alphabet is the pole, not the twist)**.** *Fix $b \neq 0$ and let $c = 1$, $d = a/b$, with a* free. *Write $\Phi'_{l,l'}$ for the coefficient of $z_1^l z_2^{l'}$ in Φ'. Then for every $f \in \Lambda$ and every N,*

$$\boxed{\sum_{l\in\mathbb{Z}} b^l\,\Phi'_{l,\,N-l}\,f \;=\; h_N\bigl[(1+b)X\bigr]\cdot f}$$

(the sum being finite, and the right side 0 for $N < 0$). The right-hand side does not involve a, whereas the two twist alphabets are $(a-b)$ and $(1-a/b)$ and both do.

Proof. Two things happen on the diagonal $z_1 = bz_2$, which is the pole of the contraction of Proposition 5.2.

The shift cancels. The total plethystic shift in Φ' is $-\frac{a-b}{z_1} - \frac{c-d}{z_2}$, i.e. $p_n \mapsto p_n - \frac{a^n - b^n}{z_1^n} - \frac{c^n - d^n}{z_2^n}$. At $z_1 = bz_2$, $c = 1$ and $d = a/b$ the bracket is

$$\frac{1}{z_2^n}\Big[-\frac{a^n - b^n}{b^n} - \big(1 - (a/b)^n\big)\Big] = \frac{1}{z_2^n}\Big[-\frac{a^n}{b^n} + 1 - 1 + \frac{a^n}{b^n}\Big] = 0$$

for every $n \geq 1$ and every a. So $f\big[X - \frac{a-b}{z_1} - \frac{c-d}{z_2}\big]$ restricted to the diagonal is $f[X]$, and f passes through untouched.

The creations double. $\sigma[Xz_1]\sigma[Xz_2]$ at $z_1 = bz_2$ is $\sigma[X\,bz_2]\sigma[Xz_2] = \sigma\big[(1+b)Xz_2\big]$, since $\sigma[A]\sigma[B] = \sigma[A+B]$ and $p_n[Xbz_2] + p_n[Xz_2] = (1+b^n)p_n z_2^n$.

Now the diagonal sum $\sum_l b^l \Phi'_{l,N-l}$ is precisely coefficient extraction of z_2^N from $\Phi'|_{z_1=bz_2}$: writing $\Phi' = \sum_{l,l'} \Phi'_{l,l'} z_1^l z_2^{l'}$ and setting $z_1 = bz_2$ gives $\sum_{l,l'} \Phi'_{l,l'} b^l z_2^{l+l'}$. The substitution is legitimate because for fixed f of degree $\deg f = D$ the shift is a Laurent polynomial with $\Phi'_{l,l'} f = 0$ unless $l \geq -D$ and $l' \geq -D$, so each coefficient of z_2^N is a finite sum (this is the finiteness argument of [3, Lem. 3.3], unchanged). Combining the two displays, that coefficient is $h_N[(1+b)X]\,f$. $\square$

Corollary 5.4 (answer to (S2): option (ii))**.** *$(1+t)$ is not a contraction constant of the t-twisted fields. The contraction constant is $\frac{z_1 - z_2}{z_1 - tz_2}$, whose alphabet is $(1-t)$ — exactly the published Necoechea–Rozhkovskaya twist — and which has a* pole*, not a value, at the point $z_1 = tz_2$ where the $(1+t)$ appears. The $(1+t)$ is $1+$ (pole location): it is the doubled creation alphabet $\sigma[Xz_1]\sigma[Xz_2]$ restricted to the diagonal. The two alphabets are functions of different data — the twist needs the zero* and *the pole, the defect needs only the pole — and they coincide in form only because the Hall–Littlewood normalisation pins the zero at $a = 1$.*

Proof. Theorem 5.3: hold the pole $b = t$ fixed and vary the zero a. The twist alphabets $(a-b)$ and $(1 - a/b)$ move; the defect alphabet $(1+b)$ does not. A quantity that is constant along a deformation cannot equal one that varies along it. $\square$

Remark 5.5 (a second, cheaper witness)**.** Inside the pinned family $a = 1$, where both alphabets are functions of the single parameter t, the same conclusion follows from $t \mapsto -t$. Equation (1) holds for every value of the parameter, so applying it at $-t$ gives

$$H_m^{-t} H_n^{-1/t} + t^{-1} H_n^{-1/t} H_m^{-t} = \big(1 + t^{-1}\big)(-t)^{-m}\, h_{m+n}\big[(1-t)X\big]\cdot,$$

verified 252/252 in §6. The defect alphabet is now $(1-t)$ and the twist alphabet is $(1+t)$: in every odd mode the two have exchanged. A pair of quantities that swap under a reparametrisation of the family is not one quantity.

Remark 5.6 (on the standing hypothesis I was asked not to build on)**.** Rick's Day 180 letter asserts at 75% stated confidence that $(1+t)X$ "is exactly the Wick constant of a t-twisted charged fermion against its dual". Corollary 5.4 refutes the assertion as stated, and Proposition 5.2 says exactly how: he identified the correct *locus* — the $(1+t)$ does live at the contraction's singularity — but not the correct *object*, since at that locus the contraction is infinite. The relationship is "$1 +$ pole of the Wick constant", not "value of the Wick constant". This is the same failure mode as `a-confirmed-operator-is-not-a-confirmed-form`: right which, wrong how.

Remark 5.7 (annotation added 2026-09-09 after reading Day 180 at source)**.** The quotation above is verbatim (Day 180 summary block, Ask 2) and the refutation in Corollary 5.4 stands unchanged. Two corrections to the *framing* of the preceding remark, both in Rick's favour, recorded rather than silently edited. (i) The "75% stated confidence" is attached to his *conclusion* — "Ask 2 (N–R

$E(-u/t)$ is the dictionary?)" — not to the Wick sentence, which is the *premise* offered as evidence for it ("the shape matches: ... *so* N–R's $E(-u/t)$ IS a plausible candidate") and which carries no hedge at all. The preceding remark therefore both understates how firmly the refuted sentence was asserted and implies a hedge that is not on it. (ii) The locus/object decomposition is *his*: his summary reads "If it is, then your $ts = 1$ vanishing hyperbola is the twist-invariance locus", and his §4 separates the locus (item 2) from the object (item 3) as numbered claims. He arrived at that split first. His §4 as a whole is moreover explicitly provisional — titled "(probably)", opening "I don't have Necoechea–Rozhkovskaya on disk", with the key equation marked "(guess, needs check)" — so no committed belief is being corrected here, only an unhedged premise. Reviewed in `clio-vega/rick-review@3416822`.

Remark 5.8 (what N–R actually says, read at source 2026-09-09)**.** Rick's §4 asks whether [4] contains $H_t(z) = E(-z/t)\psi(z)E(-z/t)^{-1}$. It does not. At source lines 392–394 the twist is *left multiplication*, and it is *asymmetric*: $\Psi^+(u) = E(-u/t)\Phi^+(u)$ but $\Psi^-(u) = H(u/t)\Phi^-(u)$ — not a twist and its inverse. The twisted anticommutator (Prop. `ferm_twisted`, ll. 398–405) is $(1 - vt/u)\Psi^+(u)\Psi^-(v) + (1 - ut/v)\Psi^-(v)\Psi^+(u) = \delta(u, v)(1 - t)^2$: the constant is $(1 - t)^2$, and the relation is single-parameter. The two-parameter exchange relation is Proposition 5.2 above, at $a = c = 1$, $b = t$, $d = s$.

6 Verification

All primary computations for this paper were run on two engines written this session: `clio-vega/proofs@69f93c6:` (Maya sets and bead moves only; no Young diagrams, no symmetric functions) and `.../vertex.py` (power-sum dictionaries; no h-basis, no abacus); that directory's `README.md` tabulates which script establishes which line below. The two engines of the 8 September self-review, `clio-vega/rick-review@314f24a:20` (border strips on Young diagrams) and `hbasis.py` (h-basis dictionaries, no power sums), were used only as *cross-checks* and share no code path with either primary.

1. **Primary engine calibrated against a proved theorem.** All one-bead matrix elements of $[R_e(t), R_f(s)]$ at generic symbolic (t, s), for $|\lambda| \leq 6$ and $1 \leq e, f \leq 4$: 1232/1232 agree with Proposition 2.1; 597 two-bead entries seen. Untuned anchor: $[R_1, R_2]s_\varnothing = (1 + t)s_{(2,1)}$, recomputed from the abacus and matching [1, §3].

2. **Theorem 2.3, closed form.** 2588/2588 over the same range at $s = 1/t$ — *every* legal $(e+f)$-move, not only the nonzero ones. The split is 1232 nonzero and 1356 zero, and the predicate $\kappa = 0$ separates them with 0 failures in either direction. κ realises every value in $\{-4, \ldots, 6\}$, so the statement is not vacuous in its own parameter.

3. **Theorem 2.3(ii), order.** All 1232 nonzero elements have order of vanishing exactly 1 at $t = -1$; the histogram is $\{1 : 1232\}$ with no other order occurring. The leading coefficient test $\Xi'(-1) = (-1)^N\kappa$ passes 1232/1232.

4. **Two-bead sector on the hyperbola.** 0 of the 597 generically-nonzero two-bead entries survive at $s = 1/t$, confirming [2, Cor. 4.2(iii)] independently. (We do not use [2, Cor. 4.2(iv)], which carries a known correction for $e \leq 2$ recorded in the annotation to that paper.)

5. **Cross-check of Theorem 2.3 on `ribbon.py`.** 764/764 one-bead elements of order exactly 1, with the same leading coefficient $(-1)^N\kappa$ and 0 failures. That engine computes R_g by enumerating border strips on Young diagrams and knows nothing of κ.

6. **Lemma 3.1.** The power-sum route ($h_N = \sum_\mu p_\mu/z_\mu$ then $p_k \mapsto (1+t^k)p_k$) and the creation route ($\sum_k t^{N-k} h_k h_{N-k}$) agree coefficientwise for $N = 0, \dots, 8$.

7. **Theorem 3.4.** $\operatorname{ord} h_N[(1+t)X] = N \bmod 2$ for $N = 0, \dots, 10$, with the per-partition ladder $\{N \bmod 2, \dots, N\}$ realised in full each time. All 513 nonzero Schur matrix elements with $N \le 6$ and $|\lambda| \le 4$ are palindromic (0 exceptions), and their order histogram is $\{0, 2, 4, \dots\}$ for N even and $\{1, 3, 5, \dots\}$ for N odd: *no matrix element of even N has order* 1. Explicit order-0 witnesses at $N = 2$: $\langle s_{(2)} | \cdot | s_\varnothing \rangle = 1 + t + t^2$ and $\langle s_{(1,1)} | \cdot | s_\varnothing \rangle = t$.

8. **Cross-check of Theorem 3.4 on `hbasis.py`.** Recomputing the defect (1) from the h-basis engine for $0 \le m, n \le 4$ gives coefficient orders $0, 1, 0, 1, 0, 1, 0, 1, 0$ for $m + n = 0, \dots, 8$: exactly $N \bmod 2$.

9. **Equation** (1) **re-derived.** Theorem [3, 3.4] was recomputed from scratch on `vertex.py`: 252/252 at $u = t$ ($|\mu| \le 3$, $-2 \le m, n \le 3$), of which 182 have nonzero defect. The same identity at $u = -t$: 252/252, giving Remark 5.5.

10. **Proposition 5.2** with *four* independent symbols a, b, c, d: $64 + 64$ agreements, 0 mismatches.

11. **Theorem 5.3**, the varying test: with b symbolic and a a free symbol, the diagonal sum equals $h_N[(1+b)X]$ in 35/35 cases ($|\mu| \le 3$, $-1 \le N \le 3$). Specialising $a \in \{1, -1, 2, \frac{1}{2}\}$ at fixed pole $b = t$ moves the twist alphabets to $(1-t,\ 1-t^2,\ 1-t^3)$, $(-1-t,\ 1-t^2,\ -1-t^3)$, $(2-t,\ 4-t^2,\ 8-t^3)$, $(\frac{1}{2}-t,\ \frac{1}{4}-t^2,\ \frac{1}{8}-t^3)$ while the defect alphabet stays $(1+t,\ 1+t^2,\ 1+t^3)$ throughout.

7 What is not claimed

1. **Not claimed: that the two objects are unrelated.** They are related, and Theorem 5.3 says how: both are governed by the same distinguished point. On the vertex side that point is the pole $z_1 = tz_2$ of the contraction; on the ribbon side the $-t$ of $1 - (-t)^\kappa$ plays the analogous role for the height statistic. What is refuted is that they are the *same* $(1+t)$, i.e. that one derivation transports to the other.

2. **Not claimed: a dictionary between (e, f) and $(k, m + n)$.** The brief asked whether any dictionary makes the order functions agree. The answer is that a dictionary can only do so by landing entirely in odd $m + n$ (Theorem 4.1), and no principle selects the odd half of a family that is defined for all degrees. If a dictionary is ever produced it must explain what pins that parity; I have none.

3. **Not claimed: anything about the two-bead sector off $ts = 1$.** Everything here is on the hyperbola.

4. **Not claimed: Rick's §2 residue suggestion.** Step 3 of the brief (is Theorem (1) literally a formal residue at $u = tz_2/z_1 = 1$?) was not attempted. Proposition 5.2 makes it look very likely — the pole is simple and the δ is its residue — but "looks like a residue" is precisely the kind of statement this paper exists to refuse.

5. **On the Necoechea–Rozhkovskaya citation.** I quote the twist alphabet $(1 - t^n)$ and the one-sided, two-dressing form of the twist. I do *not* claim that paper says anything about $(1 + t)$, about the hyperbola $st = 1$, or about my defect; it does not, and Corollary 5.4 is mine.

References

[1] Clio, *The cross-rank ribbon commutator*, 7 September 2026, `proofs/2026-09-07-Q92-cross-rank-commutator.tex`.

[2] Clio, *Infinite order over the sublattice subring*, 7 September 2026 (cycle 2), `proofs/2026-09-07-c2-Q96-order-over-sublattice.tex`; Theorem 4.1 and Corollary 4.2, with the annotation of 8 September 2026 to Corollary 4.2(iv).

[3] Clio, *The two-parameter Hall–Littlewood exchange relation, and what actually lives on the hyperbola* $st = 1$, 8 September 2026, `proofs/2026-09-08-Q99-two-parameter-exchange.tex`.

[4] A. Necoechea and N. Rozhkovskaya, *Boson-fermion correspondence for Hall-Littlewood polynomials revisited*, arXiv:1902.10049; LaTeX source read at source 8 September 2026, lines 385, 392–394, 398–405, 735, 764.